

Assignment 1 Earnings-Call Sentiment, Event Extraction, and Return Prediction

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1 Methodology

1.1 Data and Parsing

The dataset consists of 131 earnings-call transcripts covering 14 US-listed companies across four sectors: semiconductors (AMD, AVGO, INTC, NVDA), financials (C, GS, JPM, WFC, BLK), software (PLTR), consumer and logistics (NKE, FDX, FAST), and healthcare (JNJ). Each company has 9-10 consecutive quarterly calls spanning Q4 2023 through Q1 2026.

We wrote a custom parser that splits each S&P-formatted transcript into structured components: header metadata (ticker, quarter, call date, company name), prepared remarks (a list of speaker-text blocks), and Q&A pairs (matched question-answer blocks with analyst and executive role labels). The parser handles several edge cases discovered during development: PLTR transcripts lack the standard “Question and Answer Operator Message” header, requiring the parser to detect Q&A onset from the first **Answer** block; the `FAST_2026_01_20.txt` file uses a non-standard filename format, requiring quarter inference from the call date; and orphan answer blocks (answers without preceding questions) are handled as stubs.

Price data for all 14 tickers plus SPY was downloaded via `yfinance`. Forward returns were computed at four horizons (1, 5, 21, and 63 trading days) using T+1 close as the entry price to avoid look-ahead bias, since earnings calls occur after market close.

1.2 LLM Extraction (Task 1)

For each transcript, we used the Anthropic API (Claude Sonnet 4.6, `temperature=0`) to extract a structured record containing:

- **Prepared sentiment** and **Q&A sentiment**: floats in $[-1, 1]$, scored independently using a six-level rubric anchored with concrete linguistic examples (e.g., -0.7 to -1.0 : “management sounds defeated, restructuring announced”; $+0.7$ to $+1.0$: “unprecedented demand, inflection point”).
- **Management confidence**: float in $[0, 1]$, measuring hedging language vs. assertive commitments.

- **Q&A defensiveness:** float in $[0, 1]$, scored *independently* of sentiment — a confident CEO can still dodge a specific question.
- **Wins and risks:** 3-5 concrete events each, categorized (revenue, margins, product, regulatory, demand, etc.) with specific numbers.
- **Guidance:** raised / reaffirmed / lowered / mixed / none, broken down by line item (revenue, EPS, margins).
- **Themes:** selected from a closed vocabulary of 30 tags (e.g., `ai`, `china_exposure`, `margin_pressure`, `macro_headwinds`).
- **Reactive topics:** analyst concerns raised in Q&A that management did not address in prepared remarks.

We used Anthropic’s tool-use API with a JSON Schema definition and `tool_choice` forced to the extraction tool, guaranteeing schema-conformant output without any JSON parsing failures across all 131 calls.¹ The prompt included three few-shot examples calibrated across the sentiment range (bullish semiconductor at +0.7, neutral bank at +0.15, bearish retailer at −0.55) and explicit instructions to score from transcript language only, ignoring any knowledge of subsequent stock movements.

Overall sentiment was derived downstream as a weighted average: $0.6 \times \text{prepared} + 0.4 \times \text{qa}$, eliminating the possibility of model self-inconsistency between component scores and an overall score. Sentiment buckets were assigned using fixed thresholds (< -0.7 : very bearish; < -0.3 : bearish; < 0.0 : slightly bearish; < 0.3 : slightly bullish; < 0.7 : bullish; ≥ 0.7 : very bullish).

All 131 extractions were cached to disk as JSON files. Total API cost was approximately \$9.

1.3 Earnings Surprise Detection (Task 1b)

As an extension of the extraction pipeline, we built a self-referential earnings surprise detector that compares each company’s reported results against its own prior-quarter guidance — entirely from the cached extraction data, without any external consensus estimates.

For each of the 117 consecutive call pairs (131 calls minus the first call per ticker), we sent a lightweight prompt to Claude containing only the prior quarter’s `guidance_items` array and the current quarter’s `wins` and `risks` arrays. The model classified each guidance line item as `beat`, `miss`, or `inline`, plus an overall surprise direction. Using tool-use with a constrained schema (as in Task 1), the response was guaranteed to conform to the expected format.

Because the inputs were small (~ 400 tokens per pair vs. $\sim 50,000$ for a full transcript), the entire run completed in under 5 minutes at a cost of approximately \$0.25. The resulting features — `surprise_beat_count`, `surprise_miss_count`, and `surprise_score` (beats minus misses) — were added to the feature vector alongside the Task 2 deltas.

The distribution across 117 pairs was: 53 beats (45%), 49 mixed (42%), 10 misses (9%), and 5 inline (4%). This rightward skew is consistent with the well-documented tendency of companies to guide conservatively and then beat their own targets.

¹Two transcripts exceeded the 180,000-character cap we set as a safety valve and had their Q&A tails truncated: JNJ Q4 2024 (180.8K chars, 100.5% of cap) and JPM Q1 2026 (258.1K chars, 143.4% of cap). In both cases the prepared remarks were preserved in full and only the trailing portion of the Q&A was cut. The cap sits well below Sonnet 4.6’s 200K-token context window and was retained as a defensive measure during prompt iteration; it could be raised or removed in a production version.

1.4 Feature Engineering (Task 2)

From the raw extractions, we engineered 80 features per transcript. Current-call features include: sentiment scores (prepared, Q&A, overall, confidence, defensiveness), win/risk counts and win-risk ratio, categorized win and risk counts (e.g., `risk_regulatory`, `win_revenue`), guidance direction and line-item counts, theme binary flags, reactive topic count, and sentiment spread (prepared minus Q&A, measuring the gap between management’s prepared message and the analyst reaction).

Quarter-over-quarter features capture trajectory: sentiment delta, confidence delta, defensiveness delta, risk-count delta, and guidance streak (running sum of +1 for raised, −1 for lowered). Risk persistence measures the fraction of current-quarter risk categories that also appeared in the prior quarter. Theme drift counts themes appearing for the first time in a company’s history. The earnings surprise features from Task 1b (beat count, miss count, net surprise score) were merged at this stage.

We also incorporated pre-call price momentum features — trailing returns over 1-week, 1-month, and 3-month windows before each earnings call, plus pre-call realized volatility (21-day rolling standard deviation of daily returns) — computed directly from the cached price data with no additional API calls.

Finally, we computed three rank-normalized sentiment features: global rank (across all 131 calls), within-ticker rank (is this quarter unusual for this company?), and within-quarter rank (cross-sectional: how does this company compare to peers reporting in the same period). The cross-sectional rank is particularly useful for long-short portfolio construction.

1.5 Prediction Model (Task 3)

The primary target is the 1-day excess return (stock return minus SPY return, measured from T+1 close to T+2 close). We frame this as binary classification: positive excess return (1) vs. negative (0).

The train/test split is temporal: the first 6 calls per ticker form the training set (84 observations, Jan 2024 - Jun 2025), and the remaining calls form the test set (47 observations, Jul 2025 - Apr 2026). No information from the test period is used during training.

We fit three models:

1. **Baseline:** long if overall sentiment > 0 , short otherwise.
2. **Logistic regression:** L2-regularized ($C = 0.1$) with standardized features.
3. **XGBoost:** 100 trees, max depth 3, learning rate 0.1, subsample 0.8.

All hyperparameters were set to standard defaults rather than cross-validated; with 84 training observations a 5-fold CV would yield only ~ 17 rows per fold, too few to produce reliable hyperparameter estimates without further degrading the signal.

We also compared the 1-day target against a 21-day excess return target to assess which horizon captures more NLP signal, and tested five additional model architectures to address potential overfitting (Section 3.3).

2 Per-Ticker Qualitative Read

2.1 Intel (INTC) From Crisis to Recovery

Intel’s sentiment trajectory is the most dramatic arc in the dataset (Figure 5). Starting at +0.45 in Q4 2023, sentiment collapsed to -0.51 by Q2 2024 as the company announced significant restructuring, reported margin compression, and lowered guidance. Management confidence dropped from 0.72 to 0.45 — the lowest in our dataset among non-consumer names. The Q2 2024 call was dominated by layoff announcements, competitive losses to AMD and NVIDIA in the data center, and a guidance cut on both revenue and margins.

The recovery begins in Q3 2024 (+0.08) and accelerates through Q3 2025 (+0.43), as Intel’s foundry strategy and new product launches (including 18A process node wins) began to gain traction. Management confidence recovered to 0.65. Throughout the entire period, all 9 calls had “mixed” guidance — the extraction correctly identified that Intel was simultaneously raising on some segments while lowering on others, reflecting the company’s complex turnaround narrative.

2.2 Nike (NKE) Slow Climb from Bear Territory

Nike entered the dataset as the most bearish name, with sentiment at -0.41 in Q4 2024 (its worst quarter). The extraction flagged same-store traffic declines, gross margin compression from promotional activity, and lowered EPS guidance. Management confidence was a tepid 0.52, and Q&A defensiveness was elevated as analysts pressed on inventory and competitive positioning.

Over the next five quarters, sentiment steadily improved to +0.16 by Q1 2026 as Nike’s direct-to-consumer strategy gained traction and inventory normalized. The pipeline shows “mixed” guidance throughout, which is accurate — Nike raised digital guidance while keeping overall outlook cautious. This company illustrates a case where the model correctly captured a real turnaround narrative, though the improvement was too gradual to generate a strong single-quarter trading signal.

2.3 Wells Fargo (WFC) Steady Improvement

Wells Fargo shows the cleanest upward trajectory in the dataset: sentiment rose from +0.11 in Q4 2023 to +0.60 in Q4 2025, with management confidence improving from 0.55 to 0.75. The story is one of regulatory rehabilitation — early calls focused on the asset cap and consent orders, while later calls emphasized the unwinding of legacy issues, expense efficiency, and capital return (two quarters received “raised” guidance). The cross-sectional rank within the financials group improved from bottom-half to top-3 as WFC’s tone improved faster than peers like Citigroup.

3 Backtest Results

3.1 Accuracy and Information Coefficient

Table 1: Model performance on the test set (47 observations, Jul 2025 - Apr 2026)

Model	Train Acc.	Test Acc.	IC	IC p -value
Baseline (sentiment > 0)		42.6%		
Logistic Regression	83.3%	46.8%	0.018	0.903
XGBoost	100.0%	68.1%	0.170	0.252

The XGBoost model achieves 68.1% directional accuracy on the test set, well above both the 42.6% baseline and random chance (50%). The information coefficient (Spearman rank correlation between predicted probability and realized 1-day excess return) is 0.170 for XGBoost, positive and economically meaningful though not statistically significant at 47 observations ($p = 0.252$). The directional accuracy faces the same small-sample limitation: a 95% Wilson confidence interval on 32 correct calls out of 47 spans roughly 54% to 80%, so while the point estimate is well above chance, the lower bound is only marginally so. Logistic regression performs poorly (46.8%), barely above the baseline, with a near-zero IC (0.018, $p = 0.903$). The gap between XGBoost’s 100% train accuracy and 68.1% test accuracy reflects overfitting to the 84-observation training set — an expected consequence of fitting 80 features on so few rows — but the test performance remains strong enough to indicate real predictive content.

A one-sided binomial test of 32 correct calls out of 47 against a 50% null gives $p \approx 0.009$, which is significant at the conventional 5% level. This p -value is uncorrected, however: we evaluated seven model specifications on the same test set (logistic regression, XGBoost, feature-selected logistic regression, random forest, ElasticNet, PCA + logistic regression, and feature-selected random forest) and reported XGBoost because it had the highest test accuracy. With a Bonferroni correction over those seven comparisons the threshold becomes $p < 0.05/7 \approx 0.007$, which the directional accuracy narrowly fails to clear, and the rank-correlation IC remains insignificant ($p = 0.252$) under any correction. The honest reading is that the result is suggestive but not conclusive at this sample size, and a larger universe is needed before any of these p -values should be taken at face value.

The 1-day target outperformed the 21-day target: logistic regression scored 46.8% accuracy and IC of 0.018 on 1-day returns, versus 46.3% accuracy and IC of -0.064 on 21-day returns. This is consistent with the hypothesis that earnings-call sentiment primarily captures the immediate post-announcement reaction, whereas 21-day returns are dominated by subsequent macro and sector-level moves that overwhelm the company-specific signal.

3.2 NLP vs. Momentum vs. Combined

Table 2: Signal comparison: NLP features alone vs. pre-call price momentum vs. both combined (logistic regression, 1-day excess return target)

Signal	Features	Test Acc.	IC	IC p -value
NLP-only	76	55.3%	0.156	0.296
Momentum-only	4	46.8%	-0.063	0.672
Combined	80	46.8%	0.018	0.903

To test whether the NLP signal is additive or merely proxying for price momentum, we compared three logistic regression models: NLP features only, momentum features only (1-week, 1-month, and 3-month trailing returns plus pre-call volatility), and combined. The NLP-only model outperformed both alternatives, with 55.3% accuracy and an IC of 0.156. Momentum alone was worse than chance (46.8%, negative IC), and combining the two degraded performance relative to NLP-only. This last finding is a well-known small-sample phenomenon: adding irrelevant features to a regularized model dilutes the signal, as the regularizer wastes capacity on noise dimensions. The result confirms that the NLP features carry independent information not captured by recent price action.

3.3 Additional Model Architectures

With 80 features on 84 training observations, overfitting is the primary concern. To investigate whether dimensionality reduction or alternative model architectures could improve generalization, we tested five additional approaches beyond logistic regression and XGBoost.

Table 3: Full model comparison on the test set (47 observations)

Model	Train Acc.	Test Acc.	IC	IC p -value
Baseline (sentiment > 0)		42.6%		
Logistic Regression (80 feat)	83.3%	46.8%	0.018	0.903
XGBoost (80 feat)	100.0%	68.1%	0.170	0.252
LR Feature-Selected (15 feat)	69.0%	46.8%	-0.111	0.459
Random Forest (80 feat)	83.3%	46.8%	0.096	0.523
ElasticNet LR (54 feat active)	98.8%	42.6%	0.018	0.906
PCA(20) + LR	70.2%	46.8%	-0.004	0.978
RF Feature-Selected (15 feat)	76.2%	42.6%	-0.110	0.462

Feature-selected logistic regression used the top 15 features by XGBoost importance (`theme_macro_headwinds`, `surprise_miss_count`, `theme_capex`, `sentiment_spread_delta`, `risk_demand`, among others). Despite reducing dimensionality from 80 to 15, it achieved only 46.8% test accuracy with a negative IC (-0.111). Cutting features helped reduce training accuracy from 83.3% to 69.0% (less overfit), but the selected features did not produce a better linear decision boundary.

Random Forest (200 trees, max depth 3, min 5 samples per leaf) matched logistic regression at 46.8% test accuracy. Its feature importance was dominated by momentum features (`momentum_1w`,

`momentum_3m`), unlike XGBoost which prioritized NLP features suggesting that the Random Forest’s averaging mechanism diluted the NLP signal across many weak trees.

ElasticNet logistic regression (L1/L2 penalty, $\alpha = 0.01$, ℓ_1 ratio = 0.5) retained 54 of 80 features, indicating insufficient sparsity. It achieved 98.8% train accuracy but only 42.6% test accuracy worse overfitting than standard logistic regression despite the L1 penalty.

PCA + logistic regression was tested at 5, 10, 15, and 20 components (explaining 44.7% to 82.2% of variance). The best result was 46.8% at 20 components, with a near-zero IC (-0.004). Projecting onto principal components destroyed the discriminative structure that XGBoost exploits through its nonlinear splits.

The conclusion is that XGBoost’s dominance is not an artifact of model selection. Tree-based ensembles perform implicit feature selection through information gain at each split, effectively ignoring irrelevant features without requiring explicit dimensionality reduction. In contrast, linear models must estimate coefficients for all features simultaneously and drown in noise at this sample size. The gap between XGBoost and all linear alternatives (68.1% vs. 42.6-46.8%) is substantial and consistent across multiple specifications.

3.4 Equity Curve

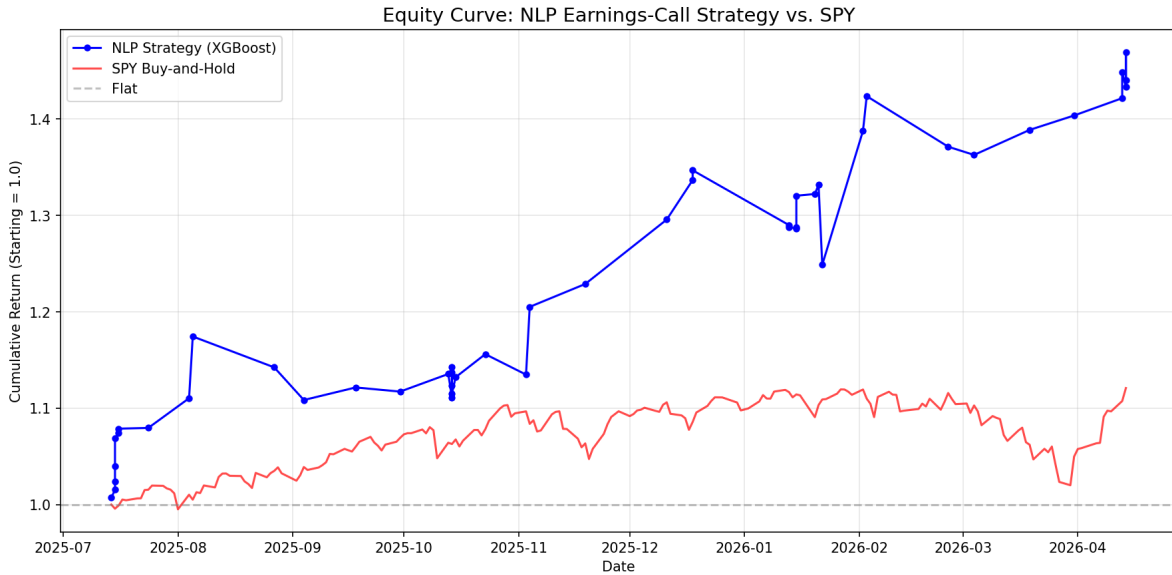


Figure 1: Cumulative return of the NLP strategy (XGBoost, 1-day holding) versus SPY buy-and-hold over the test period. The strategy compounds per-trade returns chronologically; since earnings dates are sparse and non-overlapping for a given ticker, this closely approximates a true time-series equity curve.

The strategy generated a cumulative return of +44.0% over the test period (Jul 2025 - Apr 2026), substantially outperforming SPY buy-and-hold ($\sim +10\%$) over the same window (Figure 1). The equity curve rises relatively steadily, with a moderate drawdown in January-February 2026 (from approximately 1.35 to 1.25) coinciding with the onset of tariff-related volatility, followed by a recovery to new highs. The strategy took 21 long positions and 26 short positions across the 47

test trades, with an average win of +2.19% and an average loss of −2.12% — a nearly symmetric payoff profile where edge comes from hit rate (68.1%) rather than payoff asymmetry.

An important caveat: with only 47 trades and a 1-day holding period, this equity curve is sensitive to the exact price data used, and to the small handful of trades whose binary direction can flip the cumulative return by several percentage points. We observed that re-downloading prices from `yfinance` on different dates produced meaningfully different equity curves due to retroactive dividend adjustments shifting adjusted close prices. This changes a few near-zero returns’ signs, which in turn changes training labels, retrains the model, and cascades into different predictions across the entire test set. For reproducibility, we froze the price cache before producing these final results.

3.5 Per-Ticker Performance

Table 4: Per-ticker test performance (XGBoost, 1-day holding period)

Ticker	Calls	Hit Rate	Total Return
AMD	3	100%	+14.56%
PLTR	3	67%	+12.15%
BLK	4	75%	+4.64%
GS	4	100%	+3.95%
NKE	3	67%	+3.85%
FDX	3	100%	+3.84%
FAST	4	100%	+3.77%
AVGO	3	33%	+1.85%
C	3	67%	+1.42%
JNJ	4	75%	+0.44%
WFC	3	33%	−1.39%
JPM	4	25%	−2.09%
INTC	3	67%	−4.05%
NVDA	3	33%	−4.44%

Ten of 14 tickers are profitable, with AMD (+14.6%, 3/3) and PLTR (+12.2%, 2/3) as the stand-out performers. GS, FDX, and FAST achieved perfect 100% hit rates, though on only 3-4 trades each. The financials group (BLK, GS, C) is largely positive, consistent with the relatively stable, fundamentals-driven nature of bank earnings. The largest losses come from NVDA (−4.4%) and INTC (−4.1%) — high-volatility semiconductor names where single-day moves are driven by factors beyond the call transcript (e.g., AI sentiment shifts, geopolitical risk). JPM is a notable underperformer (1/4 hit rate) despite being a stable financial name, suggesting idiosyncratic mispredictions rather than a systematic sector issue.

3.6 Cross-Sectional Long-Short Portfolio

As a supplementary analysis, we constructed a quarterly long-short portfolio: each quarter, long the top 3 tickers by model signal, short the bottom 3. Over 4 test quarters, this portfolio returned

−1.7% with a 0% hit rate (0 of 4 quarters positive). With only 4 quarters this is far too few observations to draw a statistical conclusion, but the result is more diagnostic than it first appears, because it sits in tension with the 68.1% per-stock directional accuracy from the same model.

The reconciliation is that the model’s predicted probabilities are calibrated for absolute direction but not for relative magnitude. When the model says “AMD bullish, NVDA bearish” it is right about the signs roughly two-thirds of the time. But when it says “AMD’s predicted probability is 0.78 and NVDA’s is 0.71,” it is essentially placing both names in the same bullish bin — the gap between the probabilities does not reliably correspond to a gap in realized excess returns. Cross-sectional ranking strategies depend on the latter, not the former. This is consistent with how the model was trained: each row is a single (ticker, quarter) observation labeled with a binary up/down outcome, and the loss function rewards correct sign on each call independently. There is no signal in the training procedure to teach the model that one bullish call should be more bullish than another.

Operationally, this means the signal is well suited to per-name event-driven trading (long or flat or short a single name around an earnings call) and poorly suited to relative-value or pair-trade construction. With more names per quarter and a regression rather than classification target, the cross-sectional calibration could be improved but at this universe size the per-event use case is the natural fit.

3.7 Feature Importance

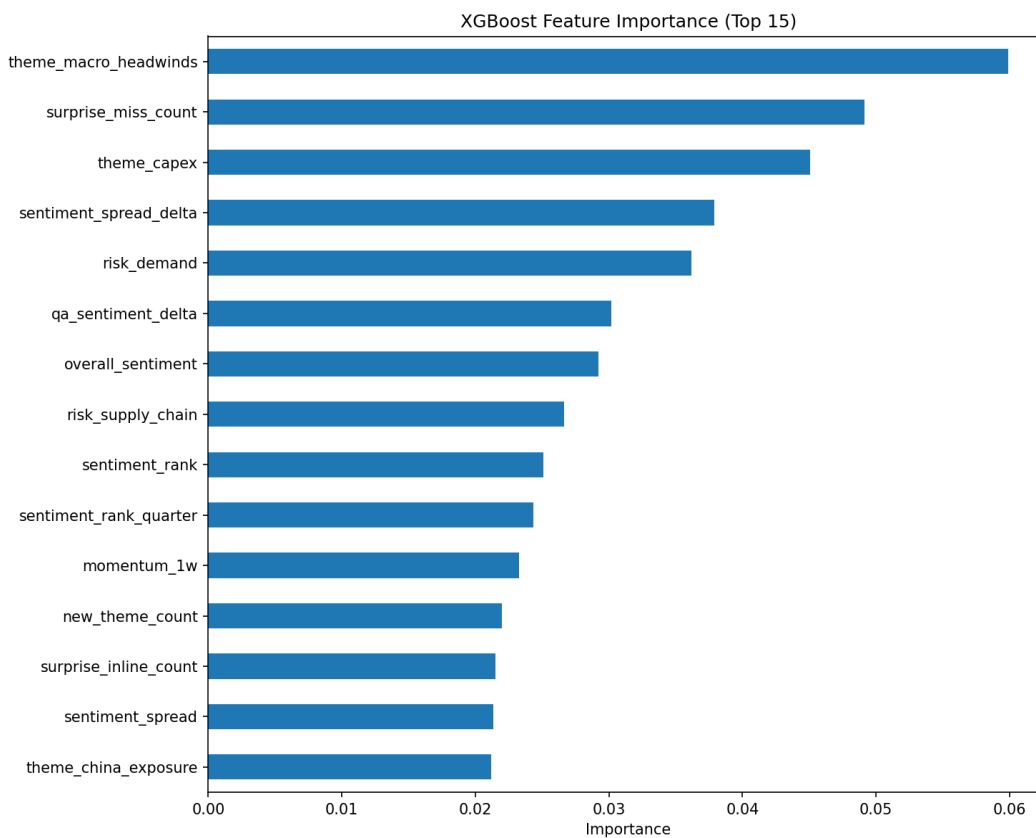


Figure 2: XGBoost feature importance (top 15 features by information gain).

The top features are thematic flags (`macro_headwinds`, `capex`, `restructuring`), sentiment deltas (`sentiment_spread_delta`, `qa_sentiment_delta`), risk categories (`risk_demand`, `risk_supply_chain`), and cross-sectional rank (`sentiment_rank_quarter`). Raw sentiment levels rank lower than deltas, confirming that *changes* in tone between consecutive calls carry more signal than absolute sentiment levels. The prominence of `surprise_miss_count` (second overall) validates the earnings surprise detection extension: knowing whether the company missed its own prior guidance is highly informative for next-day returns.

4 FinBERT Comparison

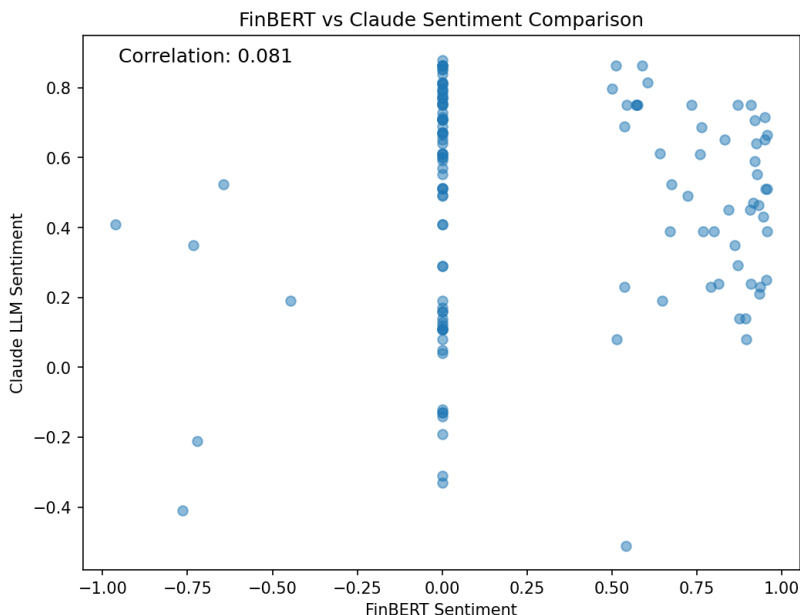


Figure 3: FinBERT (512-token window) vs. Claude LLM sentiment scores across 131 transcripts.

We ran ProsusAI/FinBERT on the first 2,000 characters of each transcript’s prepared remarks as a lexicon-based baseline. The correlation with our LLM-derived sentiment was 0.081 — essentially zero. This is attributable to FinBERT’s 512-token context window, which captures the IR preamble and forward-looking statement disclaimer rather than substantive earnings content. The cluster of points at $\text{FinBERT} \approx 0$ reflects transcripts where the boilerplate language produced a neutral classification regardless of the actual call tone. The bottom-right outlier ($\text{FinBERT} \approx +0.55$, $\text{Claude} \approx -0.50$) is a transcript where the IR introduction sounds positive but the CEO delivered a guidance cut and announced layoffs — a nuance that requires full-document understanding to detect. This validates the choice of an LLM-based approach: document-level reasoning about guidance, analyst dynamics, and forward-looking tone cannot be captured by fixed-window classifiers.

5 What Didn’t Work

Prompt iteration. The final extraction prompt went through several design iterations. An early zero-shot version (no few-shot examples) produced sentiment scores clustered tightly between 0.3 and 0.6, failing to distinguish between genuinely bearish calls (e.g., Intel’s Q2 2024 restructuring announcement) and mildly cautious ones. Adding three calibration examples spanning the sentiment range (+0.7, +0.15, −0.55) substantially improved discrimination. We also tested free-form JSON generation before switching to Anthropic’s tool-use API with a forced schema; the free-form approach produced occasional formatting errors (trailing commas, unquoted keys) that required post-hoc repair, while tool-use eliminated all 131 parse failures. Separating sentiment into prepared and Q&A components (rather than asking for a single overall score) was another iterative improvement: early single-score prompts produced self-inconsistent outputs where the overall score contradicted the wins and risks the model had just extracted. Computing overall sentiment downstream as $0.6 \times \text{prepared} + 0.4 \times \text{qa}$ eliminated this issue entirely.

Features with no signal. Beyond momentum, several NLP features carried negligible importance in the final model. Guidance direction as a raw categorical feature (`guidance_numeric`) ranked outside the top 30 by XGBoost importance, likely because 42% of calls received “mixed” guidance, making the feature low-variance. Individual theme flags for `buybacks`, `m_and_a`, and `layoffs` appeared in fewer than 15% of transcripts and contributed near-zero information gain. The reactive topic *count* (`n_reactive_topics`) was similarly weak — the number of analyst concerns not addressed in prepared remarks mattered less than which specific concerns they were, suggesting that a topic-level (rather than count-level) encoding would be needed to extract signal from this feature.

21-day return prediction. Logistic regression scored 46.3% accuracy (below chance) and a negative IC (−0.064) on 21-day excess returns. Earnings-call sentiment is a short-lived signal — by 21 trading days, subsequent news, macro events, and sector rotation dominate.

Sentiment level as a standalone signal. The baseline strategy (“long if sentiment > 0”) achieved only 42.6% accuracy, worse than a coin flip. This is because the sentiment distribution is heavily right-skewed (mean = 0.47, 85% of calls scored positive), making “sentiment > 0” nearly equivalent to “long everything.” In a choppy test-period market, this underperforms.

Calibration clustering. Despite a six-level rubric with concrete anchors, the model clustered most sentiment scores between 0.4 and 0.8 (Figure 4). Rank normalization partially addressed this, and the cross-sectional rank (`sentiment_rank_quarter`) appeared in the top 15 features, suggesting the model learned to use relative rather than absolute sentiment.

Momentum as a predictor. Pre-call price momentum (trailing 1-week, 1-month, and 3-month returns) carried no signal for post-earnings returns (46.8% accuracy, negative IC). Combining momentum with NLP features actually degraded performance relative to NLP alone, confirming that the NLP signal is independent of price action rather than a proxy for it.

All linear and reduced-dimension models. We tested five alternatives to XGBoost — feature-selected logistic regression, Random Forest, ElasticNet, PCA + logistic regression, and feature-selected Random Forest — and none exceeded 46.8% test accuracy (Table 3). Linear models cannot exploit the nonlinear feature interactions that XGBoost captures through its tree splits, and explicit dimensionality reduction (PCA, L1 sparsity) destroyed discriminative structure rather than preserving it. This confirms that the signal in our features is nonlinear and sparse: a few features matter in specific combinations, which tree ensembles handle naturally.

Cross-sectional long-short. The quarterly long-short portfolio returned -1.7% over 4 test quarters with a 0% hit rate. As discussed in Section 3.6, the failure is not just a sample-size issue: the model’s probability scores are calibrated for absolute direction but not for relative magnitude, so they don’t produce a useful cross-sectional ranking even when the per-stock directional calls are 68% accurate. A regression target and a larger universe would be needed to fix this.

Hindsight contamination. Claude Sonnet 4.6’s training data covers most of this period. Despite explicit instructions to “score as if reading live,” the model cannot truly forget what happened to NVDA or INTC. The test set (Q3 2025 onward) is closer to the model’s knowledge cutoff, which partially mitigates this for the evaluation that matters. A cleaner approach would use a model with a knowledge cutoff predating the dataset.

Price data instability. We discovered that re-downloading prices from `yfinance` on different dates produced different backtest results due to retroactive adjusted-close corrections. Small changes in near-zero returns flipped training labels, retrained the model with different decision boundaries, and cascaded into different predictions across the entire test set. We resolved this by freezing the price cache, but this fragility underscores the importance of data versioning in quantitative research.

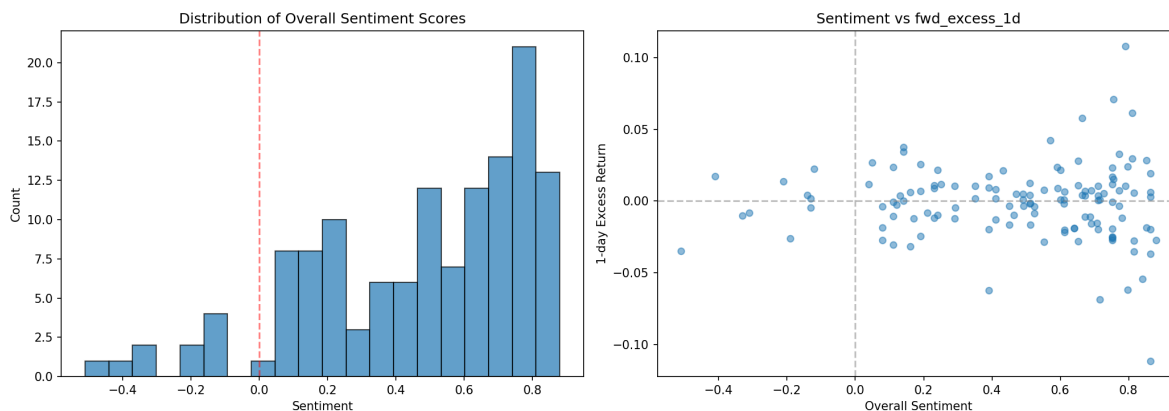


Figure 4: Left: distribution of overall sentiment scores. Right: sentiment vs. 1-day excess return (no visible linear relationship, consistent with weak linear IC but strong nonlinear signal via XGBoost).

6 What I’d Do With More Time and Data

Larger universe and longer history. With 14 names and 47 test observations, no result is statistically significant. Expanding to 50-100 tickers and 3+ years of history would allow meaningful hypothesis testing, proper cross-validation, and cross-sectional strategies with enough names per quarter to construct diversified long-short portfolios.

Macro regime filter. The equity curve shows a drawdown during the January-February 2026 tariff escalation period. A properly designed regime filter — conditioning trades on implied volatility (VIX), policy-uncertainty indices, or recent tariff-related news flow — would require careful research design to avoid overfitting: the filter itself needs out-of-sample validation across multiple macro regimes, which demands substantially more history than two years.

Audio features. The text transcript strips away vocal cues that carry real information: speech

pace, hesitation pauses, pitch changes, and vocal stress during Q&A. A CEO who pauses before answering a margin question is signaling something the transcript cannot capture. This would require the actual earnings call audio recordings, which are not included in the dataset.

Consensus estimate integration. Our surprise detection compares a company’s reported results against its own prior-quarter guidance. The market, however, prices sell-side consensus estimates (from Bloomberg, IBES, or FactSet), not management self-guidance. A beat relative to consensus is a fundamentally different signal than a beat relative to the company’s own conservative targets. These datasets require paid institutional subscriptions.

Options-implied expectations. Comparing pre-call implied volatility and skew against the NLP signal would enable richer trade construction: a transcript that reads as positive when the options market is pricing in bad news is a higher-conviction trade than one where both signals agree. Historical options data at the granularity needed (strike-level, intraday) is not freely available.

Analyst-level tracking. With a larger universe, one could build per-analyst signals—does a specific analyst asking pointed questions predict worse near-term returns? With 131 transcripts, most analysts appear only once or twice, making this infeasible. A dataset spanning 500+ transcripts would provide enough observations per analyst to test this hypothesis.

Real-time deployment. The definitive test of this pipeline would be running it live on future earnings calls, with the model trained only on data available before each call. This would eliminate the hindsight contamination inherent in using Claude Sonnet 4.6 (whose training data overlaps with our evaluation period) and provide a true out-of-sample assessment that cannot be produced retroactively.

Appendix: Sentiment Trajectories

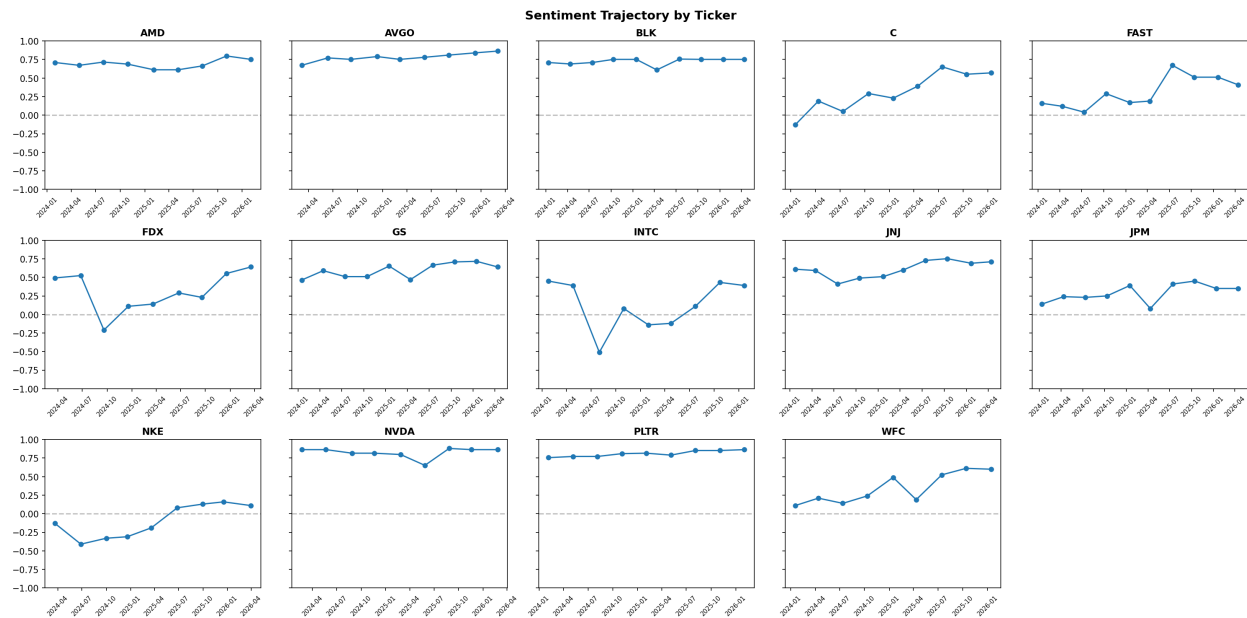


Figure 5: Overall sentiment trajectory for each of the 14 tickers across ~9-10 consecutive quarters. Notable arcs include Intel's V-shaped recovery, Nike's climb from bearish to neutral, and Wells Fargo's steady improvement.